GUIDE: Teacher-Guided Domain-Shift-Aware Learning for Remote Sensing Image Semantic Change Detection

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***Abstract*—Semantic change detection (SCD) aims to identify changed regions and their semantic categories in bitemporal remote sensing images. However, cross-temporal domain distribution shifts commonly exist between bitemporal images, making it difficult for existing methods to sufficiently perceive and learn such shifts during the encoding stage. In addition, the change branch still makes insufficient use of shallow domain distribution shift information. To overcome these limitations, this article proposes is an alias for teacher-guided domain-shift-aware learning (GUIDE), a teacher–student domain-shift-aware learning framework. First, domain distribution mixing augmentation (DDMA) is applied to the student model. It exchanges bitemporal image patches within nonchanged regions to explicitly construct domain-distribution-mixed samples, enabling the student model to perceive domain distribution shift information during both the encoding and decoding stages. Second, temporal semantic knowledge guidance (TSKG) is designed to help the student model effectively learn domain distribution shift information from domain-distribution-mixed samples. This mechanism leverages the teacher model to provide stable guidance through nonchanged region semantic supervision, cross-teacher–student contrastive learning, and cross-teacher–student temporal relation constraints. Finally, a shallow feature compensation module (SFCM) is embedded into the change branch. By dynamically fusing decoded change features with bitemporal shallow features, SFCM strengthens the utilization of shallow domain distribution shift information in the change branch. Experiments on two SCD benchmark datasets show that GUIDE outperforms existing state-of-the-art methods. Our source code is released at https://github.com/camellia-forever/GUIDE**

***Index Terms*—Data augmentation, deep learning, mean teacher, remote sensing, semantic change detection (SCD).**

I. Introduction

LAND cover continuously evolves under the combined influence of natural processes and human activities. Therefore, accurate change monitoring using remote sensing images acquired at different times has become an important research topic in Earth observation and remote sensing interpretation [1], [2], [3]. Conventional change detection mainly focuses on locating changed regions, i.e., determining "where changes have occurred." On the contrary, semantic change detection (SCD) further aims to identify the land-cover categories before and after changes and characterize the corresponding "from–to" semantic transition relationships, thereby providing more fine-grained change information at the land use/land cover level [4], [5], [6], [7], [8]. As a result, SCD can not only locate changed areas but also reveal the transition process of land-cover types, showing great application value in urban expansion monitoring, disaster assessment, resource management, and ecological environment analysis [1], [2], [3], [4].

Early SCD and related change detection methods mainly relied on handcrafted spectral, texture, color difference, or object-level features, and performed change analysis using thresholding, postclassification comparison, or traditional classifiers [9], [10], [11], [12], [13], [14], [15], [16], [17], [18]. Although these methods exhibit a certain degree of interpretability, their performance heavily depends on handcrafted features, empirical thresholds, and classifier settings, making it difficult to fully model the spatial context and cross-temporal semantic correlations in high-resolution remote sensing images. When complex land-cover distributions, imaging condition variations, and category confusion exist, traditional methods are prone to pseudo changes and semantic misclassification [1], [2], [3].

With the development of deep learning, neural network-based SCD methods have gradually become the mainstream [4], [5], [6], [7], [8], [19], [20], [21], [22], [23]. Existing methods can be roughly divided into direct classification-based, postclassification comparison-based, and multitask learning-based approaches. Direct classification-based methods regard each semantic transition relationship as an independent category for prediction, but they are easily affected by class imbalance [5], [6]. Postclassification comparison-based methods obtain change results by separately predicting the land-cover maps of the two temporal images and then comparing their categories. Although this strategy has strong semantic interpretability, classification errors can be further propagated to the final change results [21]. In comparison, multitask learning-based methods usually adopt a Siamese encoder to extract bitemporal features and construct three task branches in the decoder: two land-cover classification (LCC) branches are used to predict the land-cover categories before and after the change, while a binary change detection (BCD)

Received 1 June 2026; revised 7 August 2026; accepted 31 August 2026. Date of publication 3 September 2026; date of current version 14 September 2026. This work was supported in part by the Natural Science Foundation of Shandong Province under Grant ZR2024MF113. *(Corresponding authors: Sheng Fang; Zhe Li.)*

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Digital Object Identifier 10.1109/TGRS.2026.3730242

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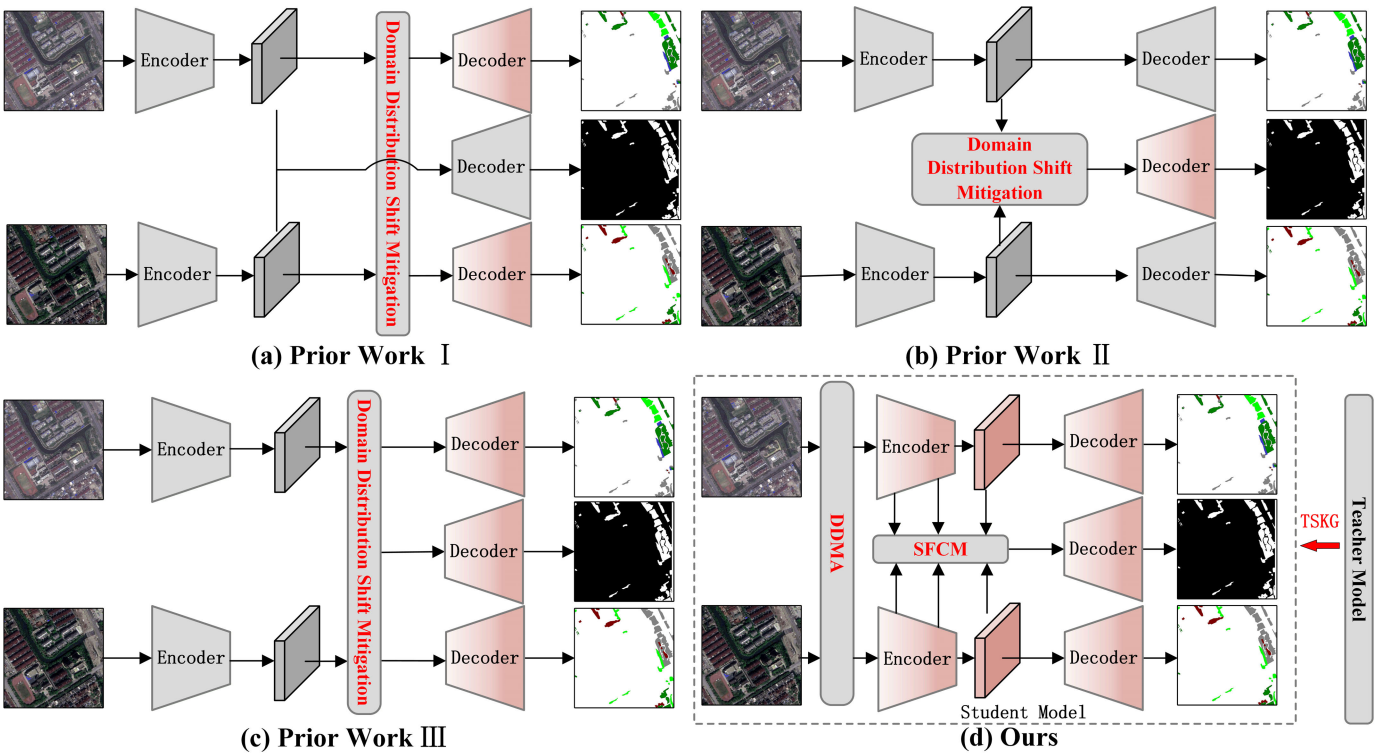


Fig. 1. Comparison between existing domain distribution shift mitigation strategies and the proposed method. (a) Semantic-branch domain distribution shift mitigation, where domain distribution shift modeling mainly acts on the bitemporal semantic decoders. (b) Change-branch domain distribution shift mitigation, where domain distribution shift modeling mainly acts on the change detection decoder. (c) Multibranch domain distribution shift mitigation, where domain distribution shift information acts on both the semantic decoders and the change decoder, but is still mainly utilized at the decoding stage. (d) Different from the above methods, the proposed method adopts a teacher–student structure for domain-distribution-shift-aware learning. DDMA constructs domain-distribution-mixed samples before feature encoding, enabling both the encoder and decoders of the student model to perceive domain distribution shift information. Meanwhile, TSKG leverages the teacher model to guide the student model in learning domain distribution shift information from domain-distribution-mixed samples, and SFCM further enhances the utilization of shallow domain distribution shift information in the change branch.

branch is employed to localize changed regions [4], [19], [23]. This triple-branch paradigm integrates change localization and semantic recognition into a unified framework, and has become a representative structure in current high-resolution SCD research [4], [8], [19], [23], [24], [25], [26], [27], [28], [29], [30], [31].

In recent years, the development of remote sensing foundation models has opened up new possibilities for change detection. Models such as HyperSIGMA and SkySense leverage large-scale pretraining to exhibit strong capabilities in general representation learning, cross-scene generalization, and knowledge transfer [32], [33]. However, general-purpose representations cannot completely replace task-specific modeling for change detection. In particular, SCD still requires fine-grained modeling of bitemporal interactions, change relationships, and semantic transitions. Therefore, the general representations provided by remote sensing foundation models still need to be integrated with task-specific mechanisms tailored to SCD [34]. Overall, foundation models provide a stronger representation basis for change detection, while specialized modeling of SCD characteristics remains an important avenue for further improving detection performance.

However, in practical bitemporal remote sensing images, images acquired at different times are often affected by various factors. These factors include acquisition time, solar elevation angle, sensor conditions, seasonal phenology, weather conditions, and changes in surface appearance. As a result, significant domain distribution shifts commonly exist between the two images. Such shifts do not necessarily correspond to real land-cover changes. However, they may alter the spectral responses, textural structures, and spatial appearances of the same land-cover objects across different temporal phases. This makes it difficult for the model to distinguish apparent changes caused by temporal shifts from real land-cover changes. Consequently, pseudochange responses are easily produced, which ultimately degrades the performance of SCD.

To address the pseudochange problem caused by cross-temporal domain distribution shifts, existing methods have attempted to alleviate the feature inconsistency between bitemporal images from different task branches. Bi-SRNet models the semantic correlations between bitemporal images through cross-temporal semantic reasoning. Its mitigation of domain distribution shifts mainly acts on the two semantic decoders, as shown in Fig. 1(a) [8]. ChangeMask strengthens change representation modeling through a temporal-symmetric Transformer, while EC-MCNet improves the discriminative capability of the change branch for temporal differences through difference enhancement and task collaboration. Both methods mainly focus on suppressing the interference of

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apparent differences caused by domain distribution shifts on change discrimination through the change detection branch, as shown in Fig. 1(b) [19], [35]. MCENet enhances the consistency between semantic branches and the change branch through the multitask consistency enhancement module and cross-task mapping connections. Therefore, its mitigation of cross-temporal domain distribution shifts affects both the two semantic decoders and the change decoder, as shown in Fig. 1(c) [36].

Although the above methods have achieved certain improvements, they still suffer from two limitations.

1) First, most existing methods alleviate cross-temporal domain distribution shifts after feature encoding. As a result, domain-shift information is mainly perceived and learned by the semantic decoders and the change decoder. Since the encoder usually processes the two temporal images separately, it lacks the opportunity to observe the two temporal domain distributions within a unified feature space. Therefore, it is difficult for the encoder to fully perceive and learn cross-temporal domain distribution shift information during the feature extraction stage.
2) Second, existing change branches still make insufficient use of shallow domain-distribution information. Under significant cross-temporal domain distribution shifts, local appearance offsets may be further amplified during change discrimination, thereby inducing more pseudochange responses.

To address the above limitations, this article proposes is an alias for guided domain-shift-aware learning (GUIDE), a teacher–student domain-distribution-shift-aware learning framework. The framework aims to promote the model to fully perceive and learn bitemporal domain distribution shift information during both the encoding and decoding stages, while strengthening the utilization of shallow domain distribution shift information in the change branch. In this way, GUIDE improves the model's robustness to cross-temporal domain distribution shifts, reduces pseudochange responses caused by domain distribution shifts, and ultimately enhances SCD performance.

First, to address the first issue, we propose domain distribution mixing augmentation (DDMA) and further design temporal semantic knowledge guidance (TSKG). DDMA exploits the prior of bitemporal semantic consistency in unchanged regions and performs cross-temporal block exchange within the unchanged regions of the input images. Although the same unchanged region may exhibit different appearance distributions across the two temporal images due to factors such as seasonal variations, illumination conditions, and imaging conditions, its land-cover semantics remain unchanged. Therefore, performing block exchange within these regions does not alter the original semantic correspondence or introduce additional real change labels. DDMA thereby constructs domain-distribution-mixed samples, allowing each training sample to explicitly exhibit cross-temporal appearance differences caused by temporal domain shifts. In this way, cross-temporal domain distribution shift information can be introduced before feature encoding. As a result, both the encoder and the decoders can perceive bitemporal domain distribution shift information, as shown in Fig. 1(d). Considering that DDMA is essentially a data augmentation strategy, we integrate it into the mean teacher framework. The domain-distribution-mixed samples are fed into the student model for training, while the teacher model provides stable guidance. Furthermore, to guide the student model to effectively learn domain distribution shift information from the domain-distribution-mixed samples, TSKG first introduces nonchanged region semantic supervision. This compensates for the lack of semantic labels in nonchanged regions and enhances the semantic discrimination capability of the student model. Then, considering that DDMA inevitably introduces feature distribution perturbations while enhancing domain distribution shift information, TSKG adopts cross-teacher–student contrastive learning to further constrain the student feature distribution. Finally, cross-teacher–student temporal relation constraints are used to preserve the bitemporal semantic relationship, where nonchanged regions should be semantically similar and changed regions should be semantically different. This improves the temporal relation modeling capability of the student model. Unlike conventional pseudolabel supervision or consistency constraints, TSKG is specifically designed for the domain-distribution-mixed samples constructed by DDMA. It not only compensates for the missing semantic labels in nonchanged regions but also provides guidance on feature distributions and imposes temporal relation constraints. Under the teacher–student framework, DDMA and TSKG work collaboratively, enabling the model to perceive and learn cross-temporal domain distribution shift information from domain-distribution-mixed samples in a more stable manner.

Second, to address the second issue, we propose a shallow feature compensation module (SFCM) and embed it into the change branch of the teacher–student model. This module dynamically fuses the decoded features of the change branch with the shallow features from the two temporal phases. By introducing shallow domain distribution shift information, SFCM strengthens the utilization of shallow domain distribution shift cues in the change branch. Consequently, it further enhances GUIDE's ability to perceive and learn cross-temporal domain distribution shifts, thereby reducing pseudochange responses caused by significant cross-temporal domain distribution shifts.

The main contributions of this article are summarized as follows.

1) We propose GUIDE, a teacher–student domain-shift-aware learning framework. This framework aims to enable the model to sufficiently perceive and learn bitemporal domain distribution shift information during both the encoding and decoding stages. It also enhances the utilization of shallow domain distribution shift information in the change branch.
2) We design DDMA, which performs patch exchange within nonchanged regions to construct domain-distribution-mixed samples. This enables the training samples to explicitly present style discrepancies caused by temporal shifts, allowing both the encoder and

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decoders to perceive bitemporal domain distribution shift information.

3) We design TSKG to provide stable guidance for the student model through nonchanged region semantic supervision, cross-teacher–student contrastive learning, and cross-teacher–student temporal relation constraints. With this guidance, the student model can effectively learn domain distribution shift information from domain-distribution-mixed samples.
4) To address the insufficient utilization of shallow domain distribution shift information in the change branch, we introduce SFCM. It dynamically fuses the decoded features of the change branch with the shallow features from the two temporal phases, thereby strengthening the utilization of shallow domain distribution shift information in the change branch.

The remainder of this article is organized as follows. Section II reviews related works. Section III presents the proposed methods in detail. Section IV reports the experimental setup, results, and analysis. Section V concludes this article.

II. Related Work

A. Semantic Change Detection

BCD aims to identify changed and unchanged regions from bi-temporal remote sensing images and constitutes a fundamental task in remote sensing change detection. With the development of deep learning, end-to-end convolutional neural networks have gradually become the dominant paradigm for BCD. Daudt et al. [37] proposed three fully convolutional networks, namely FC-EF, FC-Siam-conc, and FC-Siam-diff, providing fundamental architectures for end-to-end bi-temporal feature fusion and change prediction. Subsequently, STANet introduced a spatial–temporal attention mechanism to model long-range dependencies between bi-temporal images [38], while bitemporal image transformer (BIT) employed semantic tokens to capture contextual relationships between the two temporal images, thereby further enhancing change-related information interaction and global relation modeling [39]. To fully exploit semantic information and spatial details from different encoding stages, subsequent studies adopted dense cross-layer connections, dynamic multi-level fusion, and cross-scale feature interaction to improve multi-scale change representation and spatial detail recovery [40], [41], [42], [43]. In addition, some methods introduced exchange operations at the feature level, using spatial-region and channel exchange to facilitate information flow between the bi-temporal branches, or combining hierarchical feature exchange with channel–spatial difference modeling to strengthen change representation [44], [45]. More recently, ChangeMamba introduced state-space models into change detection, offering a new perspective for efficient global spatial–temporal modeling [46]. Meanwhile, hyperspectral change detection has also witnessed notable progress. STMNet constructs pseudo bi-temporal samples from single-temporal hyperspectral images to enable self-supervised change detection [47], whereas MambaFedCD combines Mamba with active federated learning to collaboratively model spatial, spectral, and temporal information, thereby improving hyperspectral change detection performance [48].

Although the above BCD methods have made significant progress in change-region localization, they usually only output binary change results and cannot further characterize the semantic categories of changed regions. To obtain more fine-grained "from-to" semantic change information, SCD further requires identifying both changed locations and their semantic categories before and after changes. Early SCD methods mainly focused on bi-temporal feature extraction and semantic change classification. For example, asymmetric siamese network (ASN) uses an asymmetric Siamese structure to characterize land-cover distribution differences between bi-temporal images and constructs the SECOND dataset [5]. SCDNet adopts Siamese UNet to extract multi-scale features, and combines atrous convolution, attention mechanisms, and deep supervision to improve the generation of semantic change maps [7].

Since SCD involves both change localization and bi-temporal semantic recognition, multi-task learning has gradually become an important modeling paradigm in this field. MTSCD-Net jointly models semantic segmentation and change detection under a multi-task learning framework. It also uses the location prior provided by the change detection branch to guide the semantic segmentation branch [23]. Subsequently, some methods further focus on information interaction and collaborative optimization among multiple task branches. For instance, some methods enhance multi-branch feature interaction at the decoding stage [49]. Others explicitly model the "from-to" semantic transition relationship between bi-temporal images to improve the representation of complex semantic changes [50].

Notably, unlike existing methods that perform bi-temporal exchange at the feature level to enhance feature interaction [44], [45], the proposed DDMA conducts cross-temporal image-block exchange before encoding to construct training samples containing domain distribution shift information, thereby enabling the encoder to directly perceive cross-temporal domain distribution shifts. In addition, the proposed SFCM also adopts multi-level feature fusion, but its functional objective differs from that of existing BCD methods, which primarily focus on enhancing change representation and spatial detail recovery [40], [41], [42], [43], as well as existing SCD methods, which mainly strengthen information collaboration among different branches [49]. SFCM is designed to enhance the utilization of shallow cross-temporal domain distribution shift information in the change branch, thereby improving the ability of GUIDE to perceive and learn domain distribution shifts. Specifically, SFCM adaptively integrates shallow domain distribution shift information into the change branch.

B. Mean Teacher

Mean Teacher is a representative semi-supervised learning framework, whose core idea is to provide stable supervisory signals for the student model through a teacher model. Tarvainen and Valpola proposed the Mean Teacher method, in which the teacher model is constructed using an exponential moving average (EMA) of the student model parameters, and prediction consistency between the teacher and student models

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is enforced under different input perturbations [51]. Compared with directly generating pseudo-labels using the current model, the EMA teacher generally provides smoother and more stable predictions. Building upon this idea, subsequent studies have further explored teacher–student learning and related consistency learning strategies to more effectively exploit unlabeled data.

In the remote sensing community, Mean Teacher and related consistency learning strategies have been widely applied to semi-supervised semantic segmentation. Wang et al. [52] mined potential supervisory information from unlabeled samples through consistency constraints and adopted an average pseudo-label updating strategy to improve pseudo-label stability. Cui et al. [53] proposed a dual cross-entropy consistency learning method to enhance supervision reliability by constraining the predictions of the teacher and student models. Xin et al. [54] further proposed a feature perturbation-based consistency regularization method, extending perturbations from the image level to the feature level and combining consistency learning with contrastive learning to improve model robustness against feature perturbations. Overall, these methods mainly reduce uncertainty in learning from unlabeled samples by enforcing prediction consistency across different models, branches, or perturbation conditions.

Moreover, Mean Teacher and related consistency learning strategies have also been increasingly applied to remote sensing change detection to improve the utilization of unlabeled bi-temporal images. Mao et al. [55] introduced mean teacher into remote sensing change detection, where the teacher model generates stable pseudo-labels to supervise the student model. Local–global tri-consistency (LGTC) builds a consistency learning framework upon FixMatch and introduces tri-level consistency constraints at the pixel, region, and image levels to better exploit unlabeled bi-temporal images [56]. CutMix-CD mixes bi-temporal samples together with their supervision information and further imposes teacher–student consistency constraints to enhance prediction stability on the mixed samples [57]. For SCD, SSCLNet employs semi-supervised contrastive learning to strengthen bi-temporal semantic representations, thereby improving SCD performance [58]; S^3CDRNet jointly models bi-temporal semantic information and the spatial details of changed regions to better preserve semantic information and change details [59]. Overall, these methods mainly improve the utilization of unlabeled data through pseudo-label generation, prediction consistency, semantic representation learning, and change-detail preservation.

Notably, although CutMix-CD and DDMA both adopt block exchange operations, CutMix-CD mainly performs mixing between different samples from the same temporal phase to facilitate learning from unlabeled data [57]. In contrast, DDMA exploits semantic consistency in unchanged regions and performs cross-temporal exchange within unchanged regions of the same bi-temporal image pair. In this way, it constructs training samples containing cross-temporal domain distribution shift information while preserving the original semantic relationships. In addition, TSKG is also built upon the Mean Teacher framework, but its purpose is not limited to exploiting unlabeled data through teacher-generated pseudo-labels or final prediction consistency. Instead, it guides the student model to perceive and learn the cross-temporal domain distribution shift information introduced into DDMA samples. Specifically, TSKG first uses the semantic predictions of the teacher model to supervise unchanged regions in DDMA samples. It then introduces cross teacher–student contrastive learning to constrain the stable semantic features extracted by the teacher model from the original samples and the semantic features extracted by the student model from domain-distribution-mixed samples. Unlike SSCLNet, which mainly performs contrastive learning within the student model to improve class discrimination [58], TSKG establishes cross-network contrastive relationships between the teacher and student models to constrain the student semantic feature distribution after DDMA perturbation. Finally, TSKG introduces a cross teacher–student temporal relation constraint to preserve bi-temporal semantic similarity in unchanged regions and bi-temporal semantic dissimilarity in changed regions. Therefore, the guidance provided by TSKG covers not only semantic category knowledge, but also semantic feature distributions under domain perturbations and cross-temporal semantic relationships determined by change states.

C. Domain Adaptation for Remote Sensing Images

Existing domain adaptation methods for remote sensing images typically learn transferable feature representations by reducing the distribution discrepancies between the source and target domains in the input, feature, or output space, thereby mitigating performance degradation caused by variations in geographic regions, imaging sensors, and data acquisition conditions. Zhu et al. [60] enhanced the consistency of class representations across domains through output-space adaptation and domain-level prototype memory. Bai et al. [61] jointly employed adversarial learning and contrastive learning to promote feature adaptation between the source and target domains from the perspectives of global distribution alignment and class-discriminative structure preservation.

In the field of change detection, increasing attention has been paid to the limited comparability of bi-temporal images caused by modality discrepancies. Auto-encoder kolmogorov-arnold network (AEKAN) constructs a superpixel-based AutoEncoder Kolmogorov–Arnold Network to mine latent common features between heterogeneous bi-temporal images, thereby improving the comparability of cross-modal representations [62]. Graph contrastive learning network (GCLN) constructs graph structures based on superpixels and models near–far dependencies through graph feature learning and contrastive learning, thereby enhancing the change discrimination of heterogeneous bi-temporal images [63]. Commonality graph structure learning (CGSL) learns common graph structures shared by images from different modalities to establish comparable cross-modal structural representations and mitigate the interference of modality discrepancies in change information extraction [64]. In addition, Triple-branch difference fusion network (TBDFN) adopts a triple-branch difference fusion architecture to extract and fuse difference information from heterogeneous bi-temporal images through multiple branches, thereby enhancing land-cover change representation [65].

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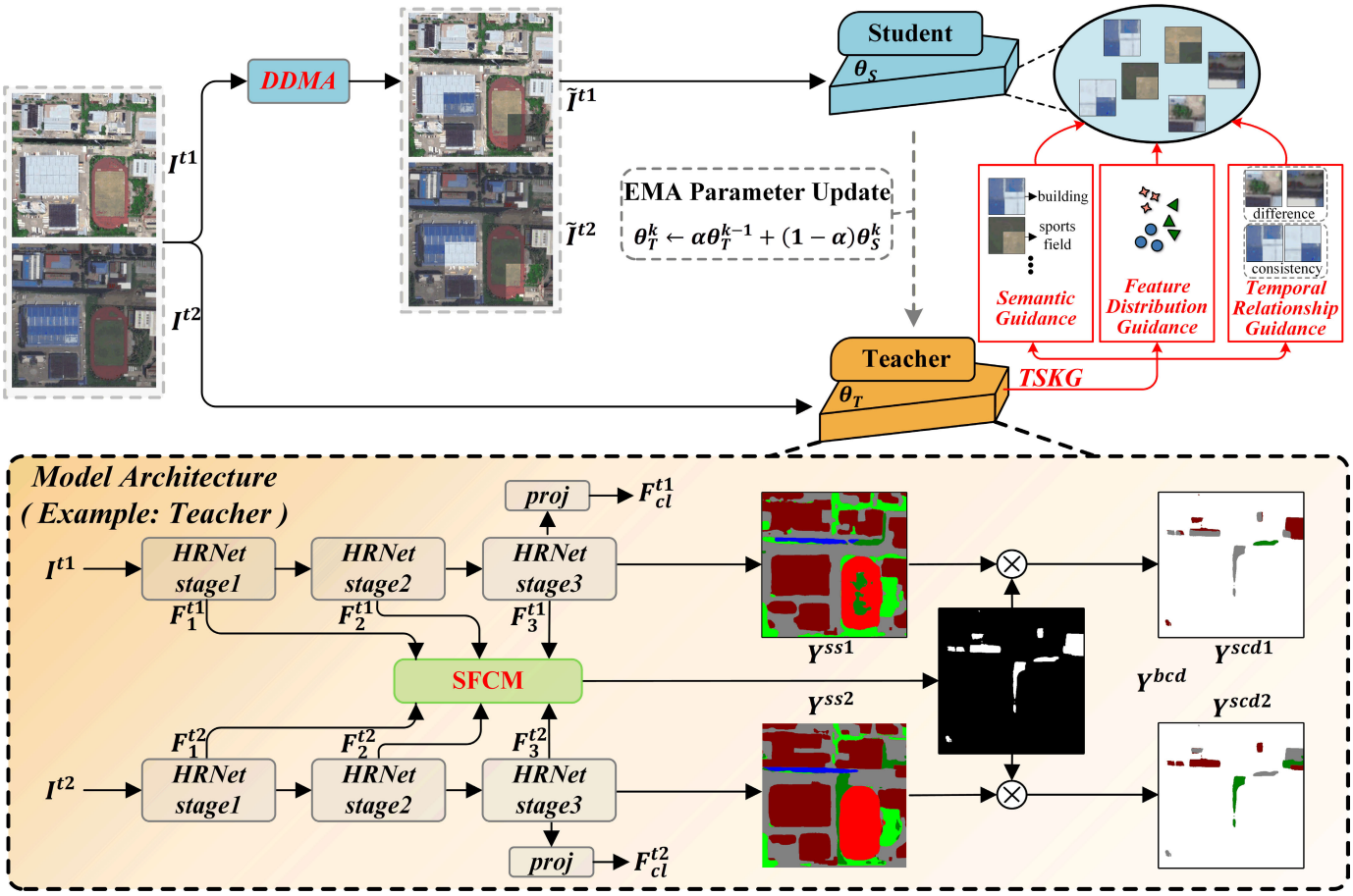


Fig. 2. Overall framework of GUIDE. The upper part illustrates the Mean Teacher-based teacher–student training framework. Given the original bi-temporal images I^{t1} and I^{t2}, DDMA performs cross-temporal image-block exchange within non-change regions to generate the augmented samples $\tilde{I}^{t1}$ and $\tilde{I}^{t2}$ containing explicit domain distribution shift information, which are then fed into the student model. In contrast, the teacher model takes the original bi-temporal images I^{t1} and I^{t2} as input. Let θ_S^k and θ_T^k denote the parameters of the student and teacher models at the kth training iteration, respectively. The teacher parameters are updated by the EMA of the student parameters according to $\theta_T^k \leftarrow \alpha\theta_T^{k-1} + (1-\alpha)\theta_S^k$, where α denotes the EMA momentum coefficient. TSKG leverages the teacher model to provide stable supervision for the student model from three aspects, including semantic guidance, feature distribution guidance, and temporal relationship guidance, thereby encouraging the student model to perceive and learn cross-temporal domain distribution shift information from the augmented samples. The lower part shows the internal network architecture adopted by both the teacher and student models, with the teacher model shown as an example. The original bi-temporal images I^{t1} and I^{t2} are separately fed into a shared-weight HRNet encoder to extract the multi-level features $\{F_1^{t1}, F_2^{t1}, F_3^{t1}\}$ and $\{F_1^{t2}, F_2^{t2}, F_3^{t2}\}$. The deep features F_3^{t1} and F_3^{t2} are used by the semantic branches to generate the bi-temporal semantic predictions Y^{ss1} and Y^{ss2}, respectively, and are also transformed by the projection heads to obtain the contrastive features F_{cl}^{t1} and F_{cl}^{t2}. The change branch incorporates SFCM to adaptively fuse the bi-temporal shallow features with the decoded change features, thereby introducing shallow domain distribution shift information and generating the binary change prediction Y^{bcd}. Finally, Y^{bcd} is used to filter Y^{ss1} and Y^{ss2}, producing the SCD results Y^{scd1} and Y^{scd2} for the two temporal phases.

In addition, some studies have more directly addressed the effects of seasonal variations, imaging-condition differences, and cross-temporal appearance inconsistencies on change detection. Kou et al. [66] progressively reduced the distribution discrepancy between images acquired in different seasons through progressive domain adaptation. Fang et al. [67] adopted content-preserving cross-temporal image translation to transform the appearance of one temporal image toward that of the other while retaining the original land-cover content as much as possible. Cheng et al. [68] suppressed interference unrelated to genuine land-cover changes in bi-temporal images through content cleansing. Overall, these methods mainly improve the comparability of bi-temporal images or features by reducing cross-temporal non-semantic discrepancies.

DDMA likewise focuses on the effects of seasonal variations, imaging-condition differences, and cross-temporal appearance inconsistencies on change detection. Unlike the above methods, DDMA addresses the difficulty of existing SCD encoders in directly perceiving cross-temporal domain distribution shifts by actively constructing training samples containing such shift information. Specifically, DDMA exchanges local image blocks within unchanged regions of the same bi-temporal image pair, enabling each individual image to simultaneously contain distribution information from both temporal phases and thereby allowing the encoder to directly perceive cross-temporal domain distribution shifts.

III. Methodology

A. Method Overview

As shown in Fig. 2, we propose GUIDE, a teacher–student domain-shift-aware learning framework. The overall framework follows the Mean Teacher structure, consisting of a student model and a teacher model with the same network architecture. Given bi-temporal remote sensing images I^{t_1} and I^{t_2}, the student model receives the domain-distribution-mixed samples constructed by DDMA for training. The teacher model

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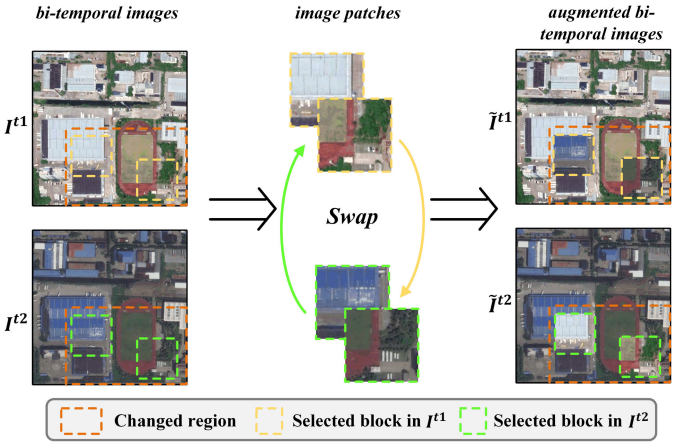


Fig. 3. Illustration of the block exchange process in DDMA. Given a pair of bi-temporal remote sensing images I^{t1} and I^{t2}, DDMA first selects image blocks at the same spatial locations within non-change regions and exchanges them between the two temporal phases, thereby obtaining the augmented bi-temporal samples $\tilde{I}^{t1}$ and $\tilde{I}^{t2}$. The orange dashed boxes indicate changed regions, while the yellow and green dashed boxes denote the image blocks selected from I^{t1} and I^{t2} for exchange, respectively. This process explicitly introduces local temporal style discrepancies into training samples without changing the original semantic change relationship.

receives the original bi-temporal images and is updated by the EMA of the student model parameters. Then, the teacher model provides stable guidance for the student model through TSKG, thereby improving the learning ability of the student model on domain-distribution-mixed samples.

Both the student model and the teacher model adopt a multi-task SCD architecture. Taking the teacher model as an example, the bi-temporal images are fed into a shared-weight HRNet encoder to extract bi-temporal features at different levels. Then, the deep semantic features are used to generate the semantic prediction results of the two temporal phases, denoted as Y^{ss1} and Y^{ss2}. The change branch is combined with SFCM to dynamically fuse the decoded change features with the bi-temporal shallow features. In this way, shallow domain distribution shift information is introduced to enhance the utilization of shallow domain distribution shift information in the change branch, and the binary change prediction result Y^{bcd} is generated. Finally, the binary change prediction result Y^{bcd} is combined with the bi-temporal semantic prediction results Y^{ss1} and Y^{ss2} to obtain the final SCD results Y^{scd1} and Y^{scd2}.

B. Domain Distribution Mixing Augmentation

To enable the model to perceive bi-temporal domain distribution shift information during the encoding stage, we propose DDMA, as shown in Fig. 3. The core idea of DDMA is to exchange bi-temporal image patches within non-changed regions. In this way, a single training sample can contain style information from different temporal phases, thereby explicitly introducing cross-temporal domain distribution shift information caused by temporal shifts at the input level. Since non-changed regions should remain semantically consistent between the two temporal phases, patch exchange within these regions does not change the original semantic change relationship. Meanwhile, it can simulate local style discrepancies caused by temporal shifts, illumination differences, and variations in imaging conditions.

Given a pair of bi-temporal remote sensing images I^{t1} and I^{t2}, and the binary change annotation G^{bcd}, we first construct a non-change mask M^{nc}

$$M^{nc}_{i,j} = \begin{cases} 1, & G^{bcd}_{i,j} = 0 \\ 0, & G^{bcd}_{i,j} = 1 \end{cases} \tag{1}$$

where $M^{nc}_{i,j} = 1$ indicates that position (i, j) belongs to the non-change region, and $M^{nc}_{i,j} = 0$ indicates that it belongs to the change region. To avoid exchanged blocks crossing real changed regions, we only select candidate blocks that are completely located within non-change regions. Given the block size b, the candidate block set Ω is defined as

$$\Omega = \left\{ (i, j) \;\middle|\; \sum_{u=0}^{b-1} \sum_{v=0}^{b-1} M^{nc}_{i+u,j+v} = b^2 \right\} \tag{2}$$

where (i, j) denotes the top-left coordinate of a candidate block. Only when all pixels in a $b \times b$ block belong to non-change regions will this block be included in the candidate set. This constraint ensures that the exchanged regions do not cover real changed regions. Thus, the original change annotation relationship is preserved.

Subsequently, DDMA randomly selects several non-overlapping candidate blocks from the candidate block set for bi-temporal exchange. To avoid excessive perturbations that may damage the original image structure, we set the maximum number of exchanged blocks K to limit the number of exchanged regions in each sample.

For each selected candidate block, DDMA performs block exchange between the bi-temporal images. Specifically, if the top-left coordinate of a selected block is (i, j), the $b \times b$ block in I^{t1} starting from (i, j) is replaced by the block at the same position in I^{t2}. Meanwhile, the corresponding block in I^{t2} is replaced by the block at the same position in I^{t1}. This process can be formulated as

$$\begin{aligned} \tilde{I}^{t1}_{i:i+b-1,\ j:j+b-1} &= I^{t2}_{i:i+b-1,\ j:j+b-1} \\ \tilde{I}^{t2}_{i:i+b-1,\ j:j+b-1} &= I^{t1}_{i:i+b-1,\ j:j+b-1}. \end{aligned} \tag{3}$$

The unselected regions remain unchanged. After this operation, the augmented bi-temporal samples $\tilde{I}^{t1}$ and $\tilde{I}^{t2}$ are obtained.

Through the above strategy, DDMA explicitly introduces cross-temporal domain distribution shift information into training samples without changing the semantic change annotation relationship. Different from methods that alleviate domain distribution shifts only on encoded features or during the decoding stage, DDMA moves domain distribution shift information forward to the input stage, enabling both the encoder and decoders to perceive bi-temporal domain distribution shift information.

C. Temporal Semantic Knowledge Guidance

To stably guide the student model to learn domain distribution shift information from the domain-distribution-mixed

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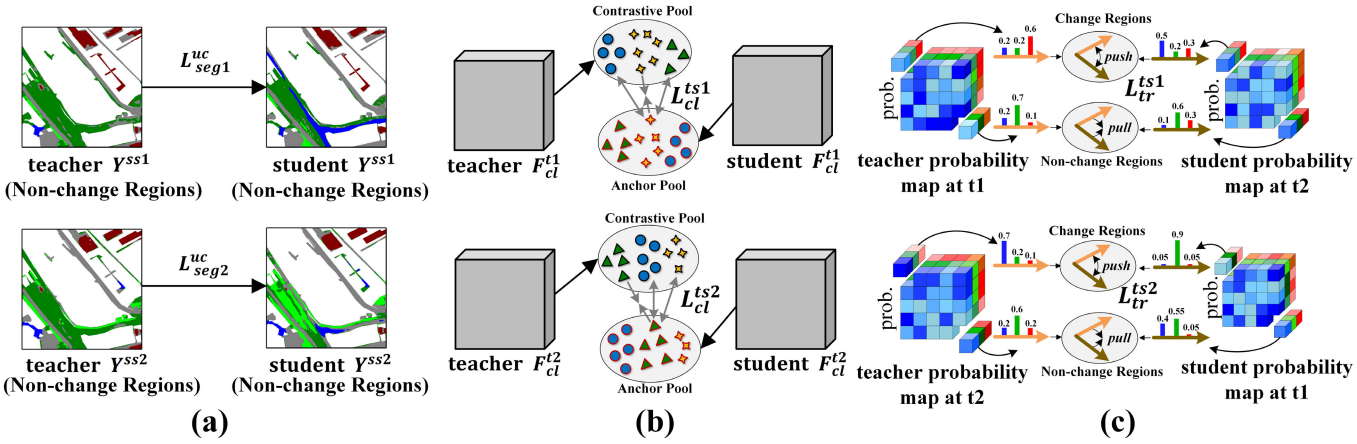


Fig. 4. Overall structure of TSKG. (a) Semantic supervision for non-change regions. The teacher model generates pseudo semantic labels for non-change regions to supervise the bi-temporal semantic predictions of the student model. Here, Y_T^{ss1} and Y_T^{ss2} denote the semantic predictions of the teacher model at the two temporal phases, while Y_S^{ss1} and Y_S^{ss2} denote the corresponding semantic predictions of the student model. L^{uc}_{seg1} and L^{uc}_{seg2} represent the non-change-region semantic supervision losses for the two temporal phases, respectively. (b) Cross-teacher–student contrastive learning. $F^{t1}_{T,cl}$ and $F^{t2}_{T,cl}$ denote the contrastive features of the teacher model at the two temporal phases, while $F^{t1}_{S,cl}$ and $F^{t2}_{S,cl}$ denote the corresponding student features. The student features form the anchor pool, whereas the teacher features form the contrastive pool. Teacher features from the same class are used as positive samples, while those from different classes are used as negative samples, producing the contrastive losses L^{ts1}_{cl} and L^{ts2}_{cl} for the two temporal phases. (c) Cross-teacher–student temporal relation constraint. The semantic prediction of the teacher model at one temporal phase is used to constrain the semantic prediction of the student model at the other temporal phase. L^{ts1}_{tr} and L^{ts2}_{tr} denote the temporal relation losses in the two cross-temporal directions, respectively. For non-change regions, the cross-temporal semantic predictions are pulled closer to preserve semantic consistency, whereas for change regions, they are pushed apart to enhance semantic discrepancy.

samples constructed by DDMA, we design TSKG. As shown in Fig. 4, TSKG consists of three components: non-changed region semantic supervision, cross-teacher-student contrastive learning, and cross-teacher-student temporal relation constraints. The teacher model generates relatively stable prediction results and feature representations from the original bi-temporal images, while the student model learns from the domain-distribution-mixed samples. With the guidance of the teacher model, the student model can improve its ability to learn cross-temporal domain distribution shift information from the domain-distribution-mixed samples in terms of semantic discrimination, feature distribution, and temporal relations.

First, considering that complete semantic labels are usually unavailable in non-change regions of some SCD datasets, we use the teacher model to provide semantic supervision for non-change regions, as shown in Fig. 4(a). Let P_T^{t1} and P_T^{t2} denote the semantic prediction probabilities of the teacher model at the two time phases, and let P_S^{t1} and P_S^{t2} denote the corresponding prediction probabilities of the student model. The teacher pseudo semantic labels are obtained by selecting the class with the maximum response

$$\hat{Y}_T^{t\tau} = \arg\max P_T^{t\tau}, \quad \tau \in \{1, 2\}. \tag{4}$$

Under the constraint of the non-change mask M^{nc}, the semantic supervision loss for non-change regions is defined as

$$\mathcal{L}^{uc}_{seg} = \frac{1}{2} \sum_{\tau\in\{1,2\}} \frac{1}{|M^{nc}|} \sum_i M_i^{nc}\, \mathcal{L}_{ce}\left(P_S^{t\tau}(i), \hat{Y}_T^{t\tau}(i)\right) \tag{5}$$

where $\mathcal{L}_{ce}$ denotes the cross-entropy loss, and i denotes the pixel position. This constraint enables non-change regions without semantic annotations to participate in training, thereby enhancing the semantic discrimination ability of the student model.

Second, we introduce cross-teacher-student contrastive learning to further constrain the feature distribution of the student model, as shown in Fig. 4(b). Specifically, the student features are used only as anchors, while the teacher features are used only as contrastive samples. Let $F^{t\tau}_{S,cl}$ and $F^{t\tau}_{T,cl}$ denote the contrastive features of the student model and the teacher model at time phase $t\tau$, respectively. For a student anchor $z^{t\tau}_{S,i}$, samples with the same class are selected from the teacher feature pool as positive samples, while samples from different classes are used as negative samples. The cross-teacher-student contrastive loss is defined as

$$\mathcal{L}^{t\tau}_{cl} = -\frac{1}{|\mathcal{A}^{t\tau}|} \sum_{i\in\mathcal{A}^{t\tau}} \log \frac{\sum_{p\in\mathcal{P}_T(i)} \exp\left(\mathrm{sim}\left(z^{t\tau}_{S,i}, z^{t\tau}_{T,p}\right)/\gamma\right)}{\sum_{q\in\mathcal{C}_T(i)} \exp\left(\mathrm{sim}\left(z^{t\tau}_{S,i}, z^{t\tau}_{T,q}\right)/\gamma\right)} \tag{6}$$

where $\mathcal{A}^{t\tau}$ denotes the student anchor set, $\mathcal{P}_T(i)$ denotes the positive sample set in the teacher feature pool, and $\mathcal{C}_T(i)$ denotes the candidate contrastive sample set. The candidate set contains both positive samples with the same class as the anchor and negative samples from different classes, namely $\mathcal{P}_T(i) \subseteq \mathcal{C}_T(i)$. $\mathrm{sim}(\cdot)$ denotes the feature similarity function, and γ is the temperature coefficient, which is set to 0.07 in this article. The cross-teacher-student contrastive loss over the two time phases is defined as

$$\mathcal{L}^{ts}_{cl} = \frac{1}{2}\left(\mathcal{L}^{t1}_{cl} + \mathcal{L}^{t2}_{cl}\right). \tag{7}$$

This constraint uses the teacher feature pool to provide stable References for student features, thereby improving the feature discrimination of the student model under domain-discrepant samples.

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Finally, we design a cross-teacher-student temporal relation constraint to encourage the student model to maintain reasonable bi-temporal semantic relations, as shown in Fig. 4(c). In SCD, the bi-temporal semantics of non-change regions should remain consistent, while the bi-temporal semantics of change regions should be different. Let $\tau \in \{1, 2\}$ denote the time-phase index, and let $\bar{\tau}$ denote the opposite time phase of τ. The cross-teacher-student temporal relation loss in the two directions can be uniformly formulated as

$$\mathcal{L}_{tr}^{ts\tau} = \frac{1}{N} \sum_{i} \begin{cases} 1 - \cos\left(P_T^{t\tau}(i), P_S^{t\bar{\tau}}(i)\right), & G_i^{bcd} = 0 \\ \cos\left(P_T^{t\tau}(i), P_S^{t\bar{\tau}}(i)\right), & G_i^{bcd} = 1. \end{cases} \tag{8}$$

The final cross-teacher-student temporal relation loss is the average of the two directions

$$\mathcal{L}_{tr}^{ts} = \frac{1}{2}\left(\mathcal{L}_{tr}^{ts1} + \mathcal{L}_{tr}^{ts2}\right). \tag{9}$$

Here, $G_i^{bcd} = 0$ denotes the non-change region. In this case, minimizing $1 - \cos(\cdot)$ pulls the cross-temporal semantic predictions closer. $G_i^{bcd} = 1$ denotes the change region. In this case, minimizing $\cos(\cdot)$ pushes the cross-temporal semantic predictions apart. This constraint improves the ability of the student model to model non-change consistency and real-change discrepancy.

In summary, the overall loss of TSKG is defined as

$$\mathcal{L}_{TSKG} = \mathcal{L}_{seg}^{uc} + \lambda_{cl}\mathcal{L}_{cl}^{ts} + \mathcal{L}_{tr}^{ts} \tag{10}$$

where λ_{cl} denotes the weight coefficient of the cross-teacher-student contrastive loss, which is set to 0.1 in this article. Through the joint guidance from the above three aspects, TSKG can fully utilize the stable predictions and feature representations of the teacher model, enabling the student model to more effectively learn the cross-temporal domain distribution shift information introduced by DDMA.

D. Shallow Feature Compensation Module

To address the insufficient utilization of shallow domain distribution shift information in the change branch, we propose SFCM. As shown in Fig. 5, SFCM dynamically fuses the decoded features of the change branch with the bi-temporal shallow features. In this way, shallow domain distribution shift information is introduced to enhance its utilization in the change branch, thereby reducing pseudo-change responses caused by significant cross-temporal domain distribution shifts.

Specifically, given the bi-temporal multi-level features extracted by the shared encoder, i.e., $\{F_1^{t1}, F_2^{t1}, F_3^{t1}\}$ and $\{F_1^{t2}, F_2^{t2}, F_3^{t2}\}$, the change branch first concatenates the highest-level bi-temporal features and applies a convolution operation to obtain the initial decoded change feature D_3^{bcd}

$$D_3^{bcd} = \mathrm{Conv}\left(\mathrm{Cat}\left(F_3^{t1}, F_3^{t2}\right)\right) \tag{11}$$

where $\mathrm{Cat}(\cdot)$ denotes channel-wise concatenation, and $\mathrm{Conv}(\cdot)$ denotes the convolution operation. To introduce shallow domain distribution shift information, we further incorporate bi-temporal shallow features into the change branch.

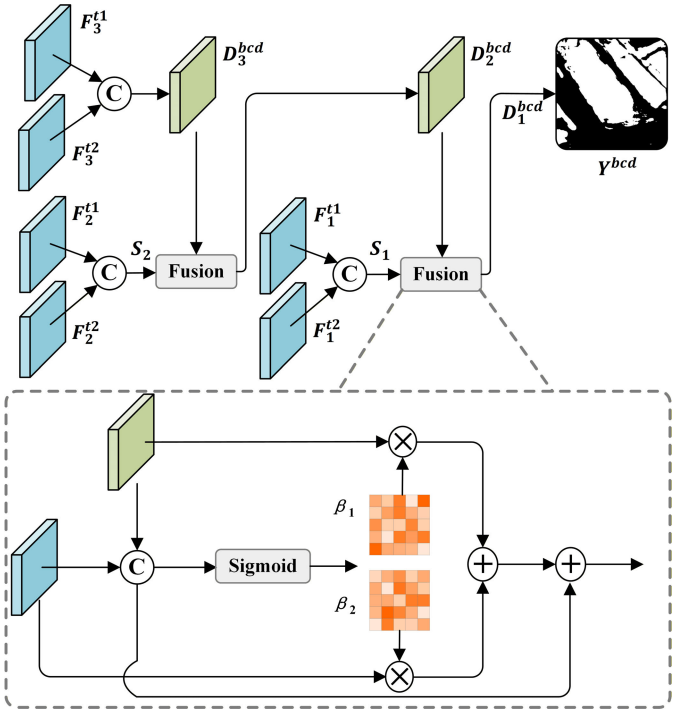


Fig. 5. Structure of SFCM. SFCM introduces shallow domain distribution shift information by dynamically fusing decoded change features with bi-temporal shallow features, thereby enhancing the use of shallow domain distribution shift information in the change branch. First, the highest-level bi-temporal features F_3^{t1} and F_3^{t2} are concatenated and transformed by a convolution operation to obtain the initial decoded change feature D_3^{bcd}. Then, the lth level bi-temporal shallow features F_l^{t1} and F_l^{t2} are concatenated and transformed by a convolution operation to obtain the corresponding shallow feature representation S_l, where $l \in \{1, 2\}$. In each fusion stage, SFCM generates dynamic fusion weights β_1^l and β_2^l based on the current decoded change feature D_{l+1}^{bcd} and the shallow feature S_l. These weights adaptively adjust the fusion ratio between the two types of features, progressively introducing shallow domain distribution shift information. Specifically, D_3^{bcd} is fused with S_2 to obtain D_2^{bcd}, and then D_2^{bcd} is fused with S_1 to obtain D_1^{bcd}. Finally, the change branch generates the binary change prediction result Y^{bcd} based on D_1^{bcd}.

For the features at the lth level, the bi-temporal features are first concatenated and then transformed by a convolution operation to obtain the corresponding shallow feature S_l

$$S_l = \mathrm{Conv}\left(\mathrm{Cat}\left(F_l^{t1}, F_l^{t2}\right)\right), \quad l \in \{1, 2\}. \tag{12}$$

Then, the Fusion module generates dynamic fusion weights based on the decoded change feature D_{l+1}^{bcd} and the shallow feature S_l. Specifically, these two features are first concatenated, and then a convolution operation followed by a Sigmoid function is used to obtain the adaptive weights

$$\left[\beta_1^l, \beta_2^l\right] = \mathrm{Sigmoid}\left(\mathrm{Conv}\left(\mathrm{Cat}\left(D_{l+1}^{bcd}, S_l\right)\right)\right) \tag{13}$$

where β_1^l and β_2^l denote the dynamic weights of the decoded change feature and the shallow feature, respectively. Based on these weights, SFCM performs weighted fusion on the two types of features and combines the convolutional transformation of their concatenation to obtain the compensated decoded change feature

$$D_l^{bcd} = \beta_1^l \odot D_{l+1}^{bcd} + \beta_2^l \odot S_l + \mathrm{Conv}\left(\mathrm{Cat}\left(D_{l+1}^{bcd}, S_l\right)\right) \tag{14}$$

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where $\odot$ denotes element-wise multiplication. Specifically, when $l = 2$, SFCM fuses D_3^{bcd} with S_2 to obtain D_2^{bcd}. When $l = 1$, SFCM further fuses D_2^{bcd} with S_1 to obtain D_1^{bcd}. Finally, the change branch generates the binary change prediction result Y^{bcd} based on D_1^{bcd}.

Through the above design, SFCM can adaptively adjust the fusion ratio between the decoded change features and the bi-temporal shallow features according to feature responses. It sufficiently introduces shallow domain distribution shift information, thereby strengthening the utilization of such information in the change branch and suppressing pseudo-change responses caused by significant cross-temporal domain distribution shifts.

E. Training Objective for the Student Model

Only the student model is optimized through gradient backpropagation, while the teacher model does not participate in backpropagation and is instead updated using the EMA of the student model parameters. Specifically, the teacher parameters are updated according to $\theta_T^k \leftarrow \alpha\theta_T^{k-1} + (1 - \alpha)\theta_S^k$, where θ_S^k and θ_T^k denote the parameters of the student and teacher models at the kth training iteration, respectively, and α denotes the EMA momentum coefficient, which is set to 0.9 in this article. Following existing SCD methods [8], [35], [49], the basic supervised losses of the student model consist of the bi-temporal semantic segmentation loss $\mathcal{L}_{ss}$, the BCD loss $\mathcal{L}_{bcd}$, and the semantic consistency loss $\mathcal{L}_{sc}$.

Specifically, $\mathcal{L}_{ss}$ adopts the cross-entropy loss to optimize the semantic prediction results of the two time phases. $\mathcal{L}_{bcd}$ adopts the binary cross-entropy loss to optimize the binary change prediction result. $\mathcal{L}_{sc}$ adopts a semantic consistency constraint based on cosine similarity, which encourages consistent bi-temporal semantic predictions in non-change regions and discriminative semantic predictions in change regions.

Combined with the proposed TSKG, the overall training objective of the student model is defined as

$$\mathcal{L}_{total} = \mathcal{L}_{ss} + \mathcal{L}_{bcd} + \mathcal{L}_{sc} + \mathcal{L}_{TSKG}. \tag{15}$$

F. Discussions

In DDMA, the block size b and the maximum number of exchanged blocks K are two key hyperparameters that affect the intensity of domain distribution shift information introduced into the augmented samples. Specifically, b controls the spatial range of each exchanged region and determines the coverage scale of local cross-temporal style discrepancies. K controls the maximum number of blocks involved in exchange for each sample and determines the spatial distribution of domain distribution shift information in the image. Therefore, b and K jointly affect the intensity and distribution of domain distribution shift information in the augmented samples.

When b is small, DDMA tends to introduce fine-grained local domain distribution shift information. However, overly small blocks may fail to form obvious cross-temporal style discrepancies. When b is large, the augmented samples can present more significant domain distribution shift information. However, overly large exchanged regions may damage local structural continuity. Similarly, a small K may lead to insufficient introduction of domain distribution shift information, while an excessively large K may cause overly strong perturbations. This may affect the model's ability to learn effective domain distribution shift information. Therefore, properly setting b and K is important for DDMA to effectively introduce domain distribution shift information.

Based on the above analysis, we conduct hyperparameter experiments on b and K in Section IV-F. These experiments are used to investigate the effect of DDMA under different intensities of domain distribution shift information and further determine appropriate parameter settings.

IV. Experiments

This section presents a systematic experimental evaluation of the proposed method. Section IV-A introduces the two SCD benchmark datasets used in the experiments, Section IV-B presents the evaluation metrics, and Section IV-C describes the implementation details. Then, Section IV-D verifies the overall effectiveness of GUIDE through comparative experiments with state-of-the-art methods. Section IV-E analyzes the contribution of each key module through ablation studies. Finally, Section IV-F further discusses the influence of key hyperparameter settings on model performance.

A. Datasets

1) Second Dataset [5]: SECOND is a widely used high-resolution remote sensing dataset for SCD. It contains 2968 pairs of bi-temporal remote sensing images with a spatial resolution of 0.5 m and an image size of 512×512. SECOND includes six semantic categories, namely buildings, low vegetation, water, bare land, trees, and playgrounds, together with one non-change category. Following the commonly used experimental setting, we divide this dataset into training and testing sets with a ratio of 4:1.

2) LevirSCD Dataset [69]: LevirSCD is a benchmark dataset for remote sensing SCD, mainly covering complex urban change scenarios in Beijing. It contains 3255 pairs of bi-temporal remote sensing images with a spatial resolution of 1–2 m and an image size of 256×256, spanning from 2010 to 2019. LevirSCD includes 16 change categories and one non-change category, covering 210 specific change types. Following the official split, we divide this dataset into training and testing sets, where the training set contains 2580 image pairs and the testing set contains 645 image pairs.

B. Evaluation Metrics

To quantitatively evaluate SCD performance, four commonly used metrics are adopted in this article, including overall accuracy (OA), mean Intersection over Union (mIoU), separation kappa (SeK), and the $F1$ score for SCD (F_{SCD}). Specifically, OA measures the overall classification accuracy across all pixels. mIoU evaluates the average overlap between the predicted regions and the ground-truth regions, which can better reflect segmentation performance under class imbalance. SeK is mainly used to measure the separability of different

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TABLE I
QUANTITATIVE COMPARISON ON SECOND AND LEVIRSCD DATASETS

Model	SECOND				LevirSCD				FLOPs (G)	Params (M)
	mIoU	SeK	F_{SCD}	OA	mIoU	SeK	F_{SCD}	OA		
HRSCD-str.3(2019)*	66.83	12.93	52.94	84.55	71.29	8.77	36.80	86.50	42.94	12.77
HRSCD-str.4(2019)*	72.07	20.16	59.45	86.72	74.12	12.31	40.91	88.36	43.96	13.71
Bi-SRNet(2022)*	72.52	21.64	61.12	87.08	74.97	18.13	47.88	89.46	190.30	23.38
DEFO-MTLSCD(2024)*	73.76	23.73	62.73	87.80	73.21	15.17	44.84	88.12	401.08	26.02
BT-HRSCD(2024)*	73.27	23.13	62.37	87.55	74.03	15.70	46.07	89.42	131.57	32.97
EC-MCNet(2025)*	73.17	22.82	62.19	87.73	74.57	18.38	49.16	89.56	352.83	27.59
STS-FINet(2025)*	72.57	22.11	61.68	87.11	74.51	19.54	50.62	89.38	245.56	33.62
SCGNet(2026)*	73.45	23.66	63.17	87.81	74.05	18.61	49.52	89.18	218.64	25.48
GUIDE(ours)	74.03	24.64	63.90	88.20	75.45	20.48	50.99	89.69	137.50	19.34

* *means it is re-implemented by us and tested under the same conditions.*

semantic categories within change regions. F_{SCD} jointly considers the precision and recall of SCD results, and reflects the comprehensive performance of the model in the SCD task.

C. Implementation Details

During training, we adopt stochastic gradient descent (SGD) as the optimizer, with a momentum of 0.9 and a weight decay of 5×10^{-4}, to improve training stability and alleviate overfitting. The model is trained for 50 epochs on both datasets, with a batch size of 8. The initial learning rate is set to 0.1 on the SECOND dataset and 0.03 on the LevirSCD dataset. For data augmentation, random rotation is adopted during training to improve the generalization ability of the model.

D. Comparative Experiments

To comprehensively evaluate the performance of the proposed method, we select several representative SCD methods for comparison on the SECOND and LevirSCD datasets, including HRSCD-str.3 and HRSCD-str.4 [4], Bi-SRNet [8], DEFO-MTLSCD [49], BT-HRSCD [24], EC-MCNet [35], STS-FINet [29], and SCGNet [27]. HRSCD-str.3 and HRSCD-str.4 are early methods that formulate SCD as a joint optimization task of semantic segmentation and change detection, and they adopt different multi-task learning strategies. Bi-SRNet models intra-temporal and inter-temporal semantic relationships through bi-temporal semantic reasoning. DEFO-MTLSCD enhances the information interaction between the semantic segmentation branch and the change detection branch from the perspective of multi-task decoder design. BT-HRSCD emphasizes the importance of high-resolution features in SCD and adopts a multi-branch decoding structure to improve spatial detail representation. EC-MCNet improves the collaboration between multi-task branches through sub-task collaboration enhancement and conflict mitigation. STS-FINet builds a spatial–temporal semantic feature interaction network to enhance semantic and spatial information interaction between bi-temporal features. SCGNet utilizes a semantic correlation guidance mechanism to model semantic relationships in bi-temporal images.

1) Quantitative Result Analysis: Table I reports the quantitative comparison between the proposed method and existing state-of-the-art (SOTA) methods on the SECOND and LevirSCD datasets. It can be observed that GUIDE achieves the best performance on both datasets. On the SECOND dataset, GUIDE obtains 74.03%, 24.64%, 63.90%, and 88.20% in terms of mIoU, SeK, F_{SCD}, and OA, respectively. Compared with the second-best results, it improves these four metrics by 0.27%, 0.91%, 0.73%, and 0.39%, respectively. On the LevirSCD dataset, GUIDE also achieves the best results, with mIoU, SeK, F_{SCD}, and OA reaching 75.45%, 20.48%, 50.99%, and 89.69%, respectively. Compared with the second-best results, the improvements are 0.48%, 0.94%, 0.37%, and 0.13%, respectively. These results demonstrate that the proposed method can consistently improve SCD performance on different datasets. In particular, GUIDE still maintains strong performance in terms of SeK and F_{SCD}, indicating that the proposed framework can improve the model's robustness to cross-temporal domain distribution shifts and enhance its discrimination capability for real semantic changes.

2) Visual Result Analysis: Figs. 6 and 7 show the visual comparisons of different methods on the SECOND and LevirSCD datasets, respectively. It can be seen that existing methods still suffer from different degrees of missed detection, false detection, and category confusion in the red-box regions. On the SECOND dataset, the comparison methods have difficulty completely identifying local changed regions in A_1 and A_2. In B_1 and B_2, they tend to produce incomplete change-region recognition and category confusion. In C_1 and C_2, they show inaccurate recognition of small changed objects. On the LevirSCD dataset, the comparison methods are unstable in recognizing elongated structural regions in A_1 and A_2. In B_1 and B_2, they show inaccurate detection of local changed objects and are prone to missed or false detections. In C_1 and C_2, local missed or false detections also occur. In contrast, the prediction results of GUIDE are more consistent with GT. GUIDE can more accurately locate changed regions and generate more accurate semantic category predictions, while reducing pseudo-change responses caused by cross-temporal domain distribution shifts.

3) Model Complexity Analysis: In addition to detection performance, Table I also reports the FLOPs and parameter numbers of different methods. To ensure a fair comparison

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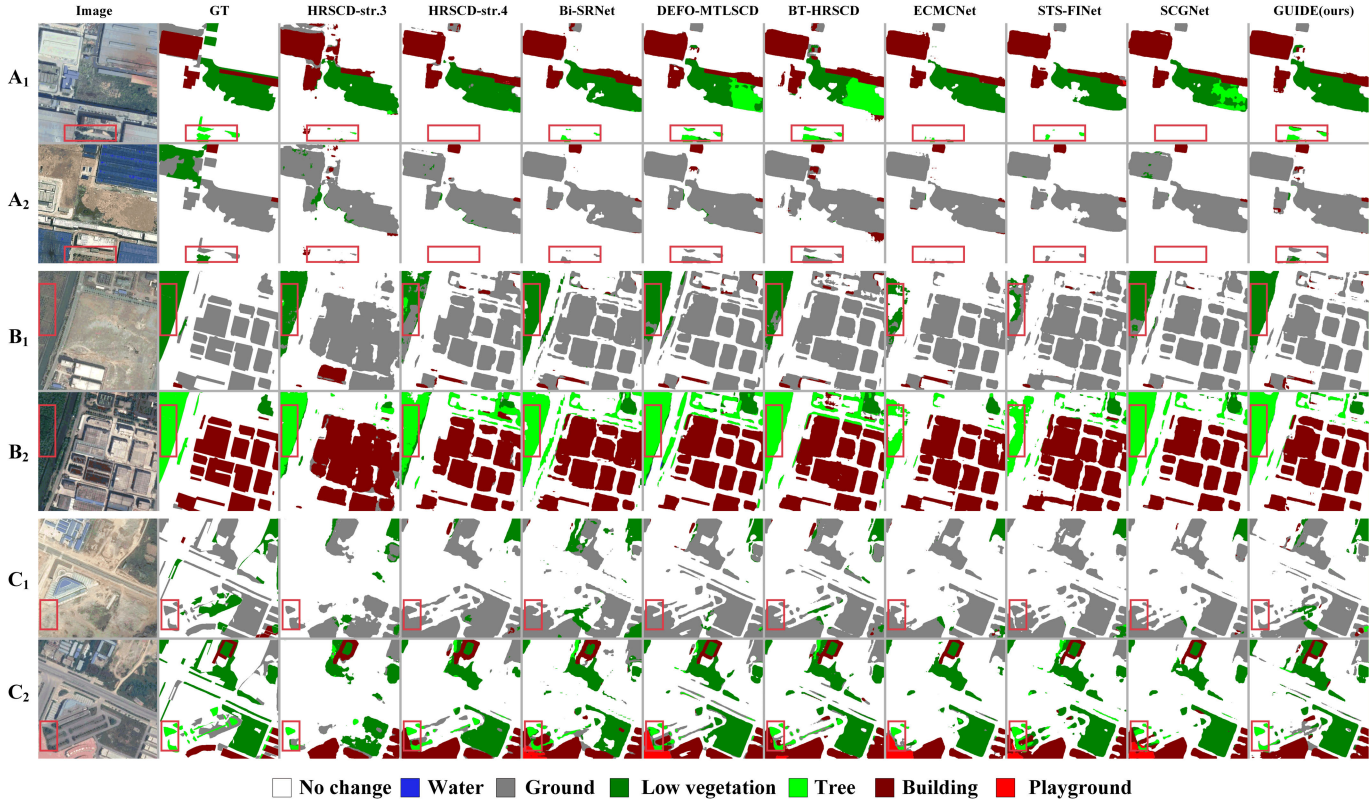


Fig. 6. Visual comparison between the proposed method and existing SOTA methods on the SECOND dataset. A_1 and A_2, B_1 and B_2, and C_1 and C_2 denote three groups of bi-temporal remote sensing images, respectively. GT denotes the ground truth. The remaining columns show the SCD results of HRSCD-str.3, HRSCD-str.4, Bi-SRNet, DEFO-MTLSCD, BT-HRSCD, EC-MCNet, STS-FINet, SCGNet, and the proposed GUIDE, respectively.

of model complexity, all methods are evaluated under the same settings, with FLOPs uniformly calculated using an input size of 512 $\times$ 512, while the number of parameters is also reported for each model. GUIDE has 137.50G FLOPs and 19.34 M parameters, indicating a moderate overall complexity. Compared with Bi-SRNet, DEFO-MTLSCD, EC-MCNet, STS-FINet, and SCGNet, GUIDE achieves better detection performance with lower computational cost or fewer parameters. Although HRSCD-str.3 and HRSCD-str.4 have lower complexity, their detection performance is significantly weaker than that of the proposed method. BT-HRSCD has FLOPs close to GUIDE, but it contains more parameters and shows lower overall performance. These results indicate that the proposed method achieves a good balance between performance and complexity.

TABLE II
ABLATION STUDY ON THE SECOND DATASET

DDMA	TSKG	SFCM	mIoU	SeK	F_{SCD}	OA
			72.90	22.87	62.28	87.33
✓			73.48	23.83	63.26	87.83
	✓		73.51	23.82	63.35	88.11
		✓	73.47	23.63	63.05	87.83
✓	✓		73.81	24.25	63.45	88.16
✓		✓	73.64	24.14	63.34	88.06
	✓	✓	73.63	23.95	63.38	88.14
✓	✓	✓	**74.03**	**24.64**	**63.90**	**88.20**

E. Ablation Study

1) Effectiveness Analysis of Each Module: To verify the effectiveness of each proposed module and their interactions, we conducted systematic module ablation experiments on the SECOND dataset, as shown in Table II. The baseline achieves mIoU, SeK, F_{SCD}, and OA scores of 72.90%, 22.87%, 62.28%, and 87.33%, respectively. On this basis, introducing DDMA alone improves the four metrics to 73.48%, 23.83%, 63.26%, and 87.83%, corresponding to gains of 0.58%, 0.96%, 0.98%, and 0.50% over the baseline, respectively. These results indicate that, by constructing domain-distribution-mixed samples, the model can be directly exposed to and learn cross-temporal domain distribution shift information, thereby enhancing its robustness. When TSKG is introduced alone, the mIoU, SeK, F_{SCD}, and OA scores reach 73.51%, 23.82%, 63.35%, and 88.11%, respectively, yielding improvements of 0.61%, 0.95%, 1.07%, and 0.78% over the baseline. Among the three individual-module settings, TSKG achieves the largest gains in F_{SCD} and OA, indicating that TSKG can provide the model with more stable and targeted supervision. When SFCM is used alone, the four metrics reach 73.47%, 23.63%, 63.05%, and 87.83%, respectively, corresponding to improvements of 0.57%, 0.76%, 0.77%, and 0.50% over the baseline. This demonstrates that further introducing shallow bi-temporal information into the change decoding stage can enhance the utilization of shallow

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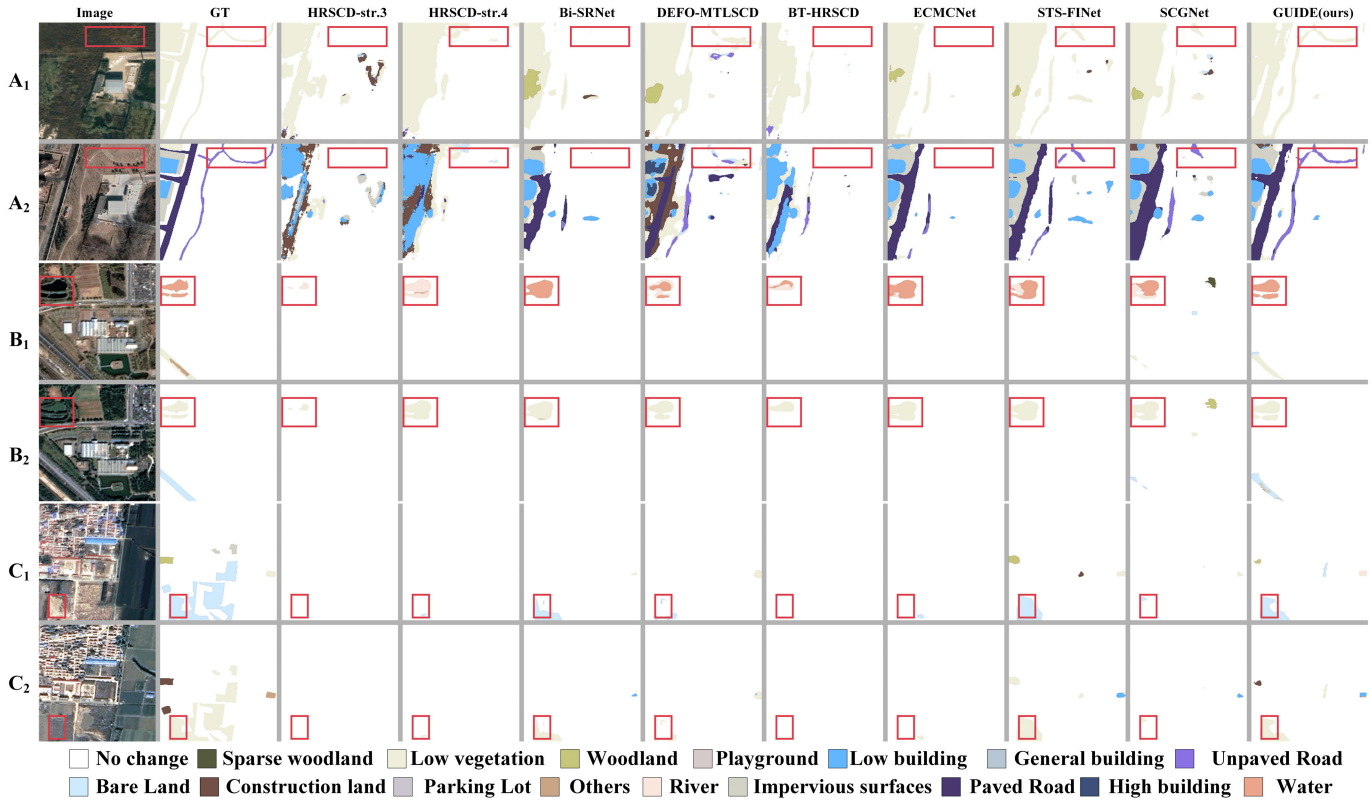


Fig. 7. Visual comparison between the proposed method and existing SOTA methods on the LevirSCD dataset. A_1 and A_2, B_1 and B_2, and C_1 and C_2 denote three groups of bi-temporal remote sensing images, respectively. GT denotes the ground truth. The remaining columns show the SCD results of HRSCD-str.3, HRSCD-str.4, Bi-SRNet, DEFO-MTLSCD, BT-HRSCD, EC-MCNet, STS-FINet, SCGNet, and the proposed GUIDE, respectively.

domain distribution shift information and thereby improve the discrimination of changed regions. It should be noted that DDMA is a strong data augmentation strategy applied to unchanged regions, where explicit semantic labels are generally unavailable. To ensure that the student model can stably learn from the domain-distribution-mixed samples generated by DDMA, the semantic supervision for unchanged regions in TSKG is retained as a basic constraint in all experimental settings involving DDMA. A check mark for TSKG in the table indicates the further introduction of the complete TSKG mechanism, which additionally includes cross-teacher–student contrastive learning and the cross-temporal relation constraint beyond the semantic supervision for unchanged regions.

Furthermore, the pairwise combination experiments demonstrate good complementarity among the proposed modules. When DDMA and TSKG are jointly employed, the mIoU, SeK, F_{SCD}, and OA scores reach 73.81%, 24.25%, 63.45%, and 88.16%, respectively, corresponding to improvements of 0.91%, 1.38%, 1.17%, and 0.83% over the baseline. This setting achieves the best overall performance among the three two-module combinations. Compared with using DDMA alone, the four metrics are further improved by 0.33%, 0.42%, 0.19%, and 0.33%, respectively, indicating that, based on the constructed domain-distribution-mixed samples, the complete TSKG can further guide the model to effectively learn the contained domain distribution shift information. In other words, DDMA increases the training samples containing real cross-temporal appearance differences, whereas TSKG mitigates the potential learning instability caused by such strongly augmented samples through multiple constraints, demonstrating strong synergy between the two modules. When DDMA and SFCM are jointly employed, the mIoU, SeK, F_{SCD}, and OA scores reach 73.64%, 24.14%, 63.34%, and 88.06%, respectively, yielding improvements of 0.74%, 1.27%, 1.06%, and 0.73% over the baseline. Compared with DDMA alone, mIoU and SeK are further improved by 0.16% and 0.31%, respectively, indicating that, while enhancing the model's perception of domain-shifted samples, strengthening the utilization of shallow domain distribution shift information in the change decoding stage can further improve change discrimination. When TSKG and SFCM are jointly employed, the four metrics reach 73.63%, 23.95%, 63.38%, and 88.14%, respectively, corresponding to improvements of 0.73%, 1.08%, 1.10%, and 0.81% over the baseline. Compared with using TSKG alone, mIoU, SeK, F_{SCD}, and OA are further improved by 0.12%, 0.13%, 0.03%, and 0.03%, respectively, demonstrating that further exploiting shallow domain distribution shift information on the basis of TSKG can further improve model performance, and that the two modules exhibit a certain degree of complementarity.

When DDMA, TSKG, and SFCM are jointly introduced, the model achieves the best performance in the table, with mIoU, SeK, F_{SCD}, and OA reaching 74.03%, 24.64%, 63.90%, and 88.20%, respectively. Compared with the baseline, these

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TABLE III
ABLATION STUDY OF THE INTERNAL COMPONENTS OF TSKG ON THE SECOND DATASET

$\mathcal{L}_{cl}^{ts}$	$\mathcal{L}_{tr}^{ts}$	mIoU	SeK	F_{SCD}	OA
		73.64	24.14	63.34	88.06
✓		73.86	24.28	63.46	88.11
	✓	73.77	24.21	63.41	88.13
✓	✓	**74.03**	**24.64**	**63.90**	**88.20**

TABLE IV
EFFECT OF THE MAXIMUM NUMBER OF EXCHANGED BLOCKS K IN DDMA ON THE SECOND DATASET

K	mIoU	SeK	F_{SCD}	OA
1	73.67	23.79	63.00	87.84
2	**74.03**	**24.64**	**63.90**	**88.20**
3	73.53	23.57	62.90	87.93
4	72.89	22.36	61.82	87.59

TABLE V
EFFECT OF THE BLOCK SIZE b IN DDMA ON THE SECOND DATASET

b	mIoU	SeK	F_{SCD}	OA
$H/4$	73.91	24.16	63.37	88.07
$H/8$	**74.03**	**24.64**	**63.90**	**88.20**
$H/16$	73.84	24.14	63.44	88.09
$H/32$	73.79	24.05	63.42	88.07

results represent improvements of 1.13%, 1.77%, 1.62%, and 0.87%, respectively, and all four metrics outperform those of every individual-module and two-module combination setting. Compared with the best-performing two-module combination, DDMA+TSKG, further introducing SFCM still improves mIoU, SeK, F_{SCD}, and OA by 0.22%, 0.39%, 0.45%, and 0.04%, respectively, further verifying the effectiveness of SFCM within the complete framework. Overall, the three modules do not simply optimize the same problem redundantly, but instead operate at different levels: DDMA explicitly constructs cross-temporal domain distribution shifts at the training-sample level, TSKG guides the model to stably learn useful information from the supervision level, and SFCM enhances the utilization of shallow domain distribution shift information in the change decoding process from the feature level. Their collaboration jointly strengthens the model's perception and learning of cross-temporal domain distribution shifts, thereby improving the overall robustness of SCD.

2) Effectiveness Analysis of the Internal Components of TSKG: To further analyze the effects of different constraints in TSKG, we conduct ablation experiments on the cross-teacher-student contrastive loss $\mathcal{L}_{cl}^{ts}$ and the cross-teacher-student temporal relation constraint $\mathcal{L}_{tr}^{ts}$. The results are shown in Table III. Since non-changed region semantic supervision serves as the basic constraint for the stable learning of DDMA samples under the Mean Teacher framework, this experiment retains non-changed region semantic supervision by default and only ablates the other two key components in TSKG. As shown in the table, when only the basic non-changed region semantic supervision is used, the model achieves 73.64% mIoU, 24.14% SeK, 63.34% F_{SCD}, and 88.06% OA. On this basis, after introducing $\mathcal{L}_{cl}^{ts}$, mIoU and SeK increase to 73.86% and 24.28%, respectively. This indicates that cross-teacher-student contrastive learning can further constrain the feature distribution of the student model, enabling it to learn domain distribution shift information from domain-distribution-mixed samples more stably. After introducing $\mathcal{L}_{tr}^{ts}$, the model performance is also improved, demonstrating that the cross-teacher-student temporal relation constraint helps preserve the bi-temporal relationship in which non-changed regions are semantically similar and changed regions are semantically different. Finally, when $\mathcal{L}_{cl}^{ts}$ and $\mathcal{L}_{tr}^{ts}$ are used together, the model achieves the best results. This suggests that the feature distribution constraint and the temporal relation constraint are complementary and jointly enhance the ability of the student model to learn cross-temporal domain distribution shift information.

F. Discussions

To further verify the influence of key hyperparameters in DDMA on model performance, we conduct experiments on the maximum number of exchanged blocks K and the block size b on the SECOND dataset. The results are shown in Tables IV and V, respectively. Here, H denotes the height of the input image.

First, as shown in Table IV, the model achieves the best performance when $K = 2$, with mIoU, SeK, F_{SCD}, and OA reaching 74.03%, 24.64%, 63.90%, and 88.20%, respectively. When $K = 1$, the performance is lower than the best result, indicating that too few exchanged blocks may introduce insufficient domain distribution shift information and thus limit the model's ability to perceive cross-temporal style discrepancies. As K further increases to 3 or 4, the model performance decreases. In particular, when $K = 4$, all metrics show an obvious decline. This indicates that too many exchanged blocks may introduce excessive perturbations and affect the model's learning of effective domain distribution shift information. Therefore, a moderate number of exchanged blocks is more beneficial for DDMA. Second, Table V shows the influence of different block sizes b on model performance. When $b = H/8$, the model achieves the best results. In contrast, when $b = H/4$, the exchanged region is relatively large. Although it can introduce more obvious cross-temporal style discrepancies, it may affect local structural continuity, leading to a slight performance decrease. When $b = H/16$ or $b = H/32$, the exchanged region is relatively small, and the model performance is also lower than that with $b = H/8$. This suggests that overly small block sizes may fail to introduce sufficiently significant domain distribution shift information. Overall, $b = H/8$ achieves a better balance between the strength of local style discrepancies and the preservation of spatial structure. In summary, K and b have an important influence on the effectiveness of DDMA. Weak block exchange may fail to sufficiently introduce cross-temporal domain distribution shift information, while overly strong block exchange may introduce excessive perturbations. The experimental results are

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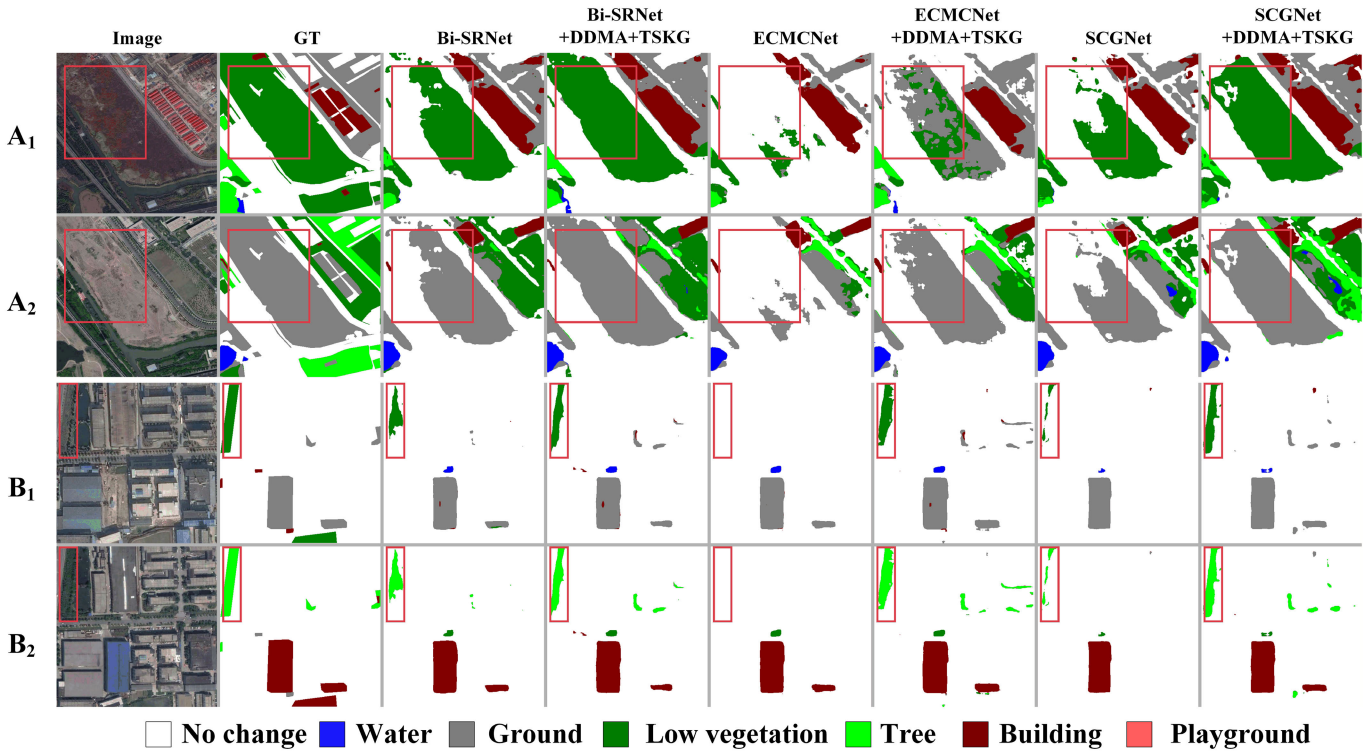


Fig. 8. Visual comparison before and after integrating DDMA and TSKG into different SCD models on the SECOND dataset. A_1 and A_2, and B_1 and B_2 denote two groups of bi-temporal remote sensing images, respectively. GT denotes the ground truth. The remaining columns show the SCD results of Bi-SRNet, Bi-SRNet+DDMA+TSKG, EC-MCNet, EC-MCNet+DDMA+TSKG, SCGNet, and SCGNet+DDMA+TSKG, respectively.

TABLE VI
TRANSFERABILITY STUDY OF DDMA AND TSKG ON DIFFERENT SCD MODELS ON THE SECOND DATASET

Model	mIoU	SeK	F_{SCD}	OA
Bi-SRNet	72.52	21.64	61.12	87.08
Bi-SRNet+DDMA+TSKG	**73.02**	**22.56**	**61.94**	**87.46**
EC-MCNet	73.17	22.82	62.19	87.73
EC-MCNet+DDMA+TSKG	**73.57**	**23.62**	**62.97**	**88.05**
SCGNet	73.45	23.66	63.17	**87.81**
SCGNet+DDMA+TSKG	**73.62**	**23.92**	**63.20**	87.78

consistent with the analysis of DDMA hyperparameters in Section III-F. That is, moderate exchange numbers and block sizes can more effectively construct domain-distribution-mixed samples, thereby improving the model's ability to perceive and learn cross-temporal domain distribution shift information. Therefore, we finally set $K = 2$ and $b = H/8$ in this article.

Based on the determined parameter settings of DDMA, we further verify the transferability of DDMA and TSKG across different network architectures. Specifically, DDMA and TSKG are integrated into different SCD models, and the performance before and after integration is compared. The results are shown in Table VI. It can be observed that after introducing DDMA and TSKG into Bi-SRNet, EC-MCNet, and SCGNet, most metrics are improved. For Bi-SRNet, after adding DDMA and TSKG, mIoU, SeK, F_{SCD}, and OA increase from 72.52%, 21.64%, 61.12%, and 87.08% to 73.02%, 22.56%, 61.94%, and 87.46%, respectively. For EC-MCNet, the four metrics are improved to 73.57%, 23.62%, 62.97%, and 88.05%, respectively. For SCGNet, mIoU, SeK, and F_{SCD} are further improved after introducing DDMA and TSKG. Although OA shows a slight fluctuation, the overall SCD performance is still enhanced. Fig. 8 further presents the visual comparison before and after module transfer. It can be observed from the red-box regions that the original models still suffer from incomplete change-region recognition, category confusion, or pseudo-change responses in some complex areas. After adding DDMA and TSKG, the prediction results become more consistent with GT, with more accurate change-region localization and more stable semantic category prediction. The above quantitative and visual results show that DDMA and TSKG do not depend on a specific network architecture. They can be transferred to different SCD models as plug-and-play modules and consistently improve the model's ability to perceive and learn cross-temporal domain distribution shift information.

V. CONCLUSION

This article addresses the cross-temporal domain distribution shift problem in remote sensing SCD and proposes GUIDE, a teacher-student domain-shift-aware learning framework. The framework follows the idea that the more sufficiently the model perceives and learns cross-temporal domain

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distribution shift information, the more robust its feature representations become. To enable the student model to perceive more sufficient domain distribution shift information, DDMA is proposed to explicitly construct domain-distribution-mixed samples containing style information from both temporal domains. Furthermore, TSKG is introduced to enable the student model to stably learn such samples under the guidance of the teacher network. In addition, SFCM is designed to strengthen the utilization of shallow domain distribution shift information in the change branch. Experimental results on the SECOND and LevirSCD datasets show that GUIDE achieves superior performance over existing advanced methods, verifying the effectiveness of explicitly constructing domain distribution shift information, stably learning such information through teacher guidance, and utilizing shallow domain distribution shift information. Future work will focus on exploring more flexible strategies for generating and learning domain-distribution-mixed samples, so as to improve the applicability and scalability of the proposed framework in broader remote sensing scenarios.

ACKNOWLEDGMENT

The authors sincerely thank the editor and the reviewers for their careful evaluation of this work. Their comments and suggestions have helped improve this article and enhance its overall quality.

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